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Modeling, simulation and techno-economic evaluation of a micro-grid system based on steam gasification of organic municipal solid waste

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ABSTRACT

The annual increase in population has led to rapid industrialization, urbanization, and economic growth over the years, becoming the lead cause of increased municipal solid waste (MSW) generation worldwide. In this study, MSW of 50 kg sampled by quartering method from Dandora dumpsite in Nairobi, Kenya is physically characterized and a sample of organic waste taken through proximate and ultimate analyses. Data obtained is used as input to a gasification model developed using advanced simulation and process engineering (Aspen) Plus software to provide characteristics of the yielded syngas. The obtained results are used to design a real gasification system and evaluate its techno-economic performance using HOMER® Pro as the tool for analysis. The techno-economic feasibility of the designed 60 kW gasification system is assessed considering a 165.59 kWh/day community-based load demand, with 11.5 % self-electricity consumption and 88.5 % grid feed. Results show organic and inorganic waste contents of 64.2 % and 35.8 %, respectively. The proximate and ultimate analyses reveal a high volatile matter content of 76.10 %, as well as fixed carbon content of 17.03 % in the biomass- rich-organic MSW. The simulated results show good agreement with experimental data and other simulation findings in literature. The study's simulation output reveals 61.15 % H_2, 1.42 % CH_4, 24.12 % CO, 13.30 % CO_2 contents following steam gasification at 700 °C. At 100 kg/h waste charge rate, a syngas yield of 2.39 m^3/kg, gasification ratio of 2.2705 kg/kg and a lower heating value of 10,150 kJ/Nm^3 (24.26 MJ/kg) are obtained. The sensitivity analysis results show increasing trend in CO_2 and H_2 syngas constituents from 0.4 to 1.0 steam-to-biomass ratio increase at 900 °C constant temperature: CO decrease is observed while CH_4 values remain negligible and unaffected. The techno-economic findings reveal that the grid sales serve to offset the system's annual costs by 39.34 %, obtaining a net present value (NPV) of $1.43 M, levelized cost of electricity (LCoE) of 0.155 $/kWh and 93.5 kg/yr. CO_2 carbon footprint. An Organic MSW steam gasification micro grid is thus financially feasible in grid connected mode and is 99 % more carbon neutral compared to the Kenyan grid.

1. Introduction

Presently, the world's population is estimated to be eight billion. This is an increase of 1.24 % compared to population in 2022 [1]. Increase in population has led to rapid industrialization, urbanization, and economic growth over the years, becoming the lead cause of increased municipal solid waste (MSW) generation worldwide. MSW handling is a problem for most countries around the world and varies with a country's economic, political and social status. The difficulty of management is even greater for developing nationsClick or tap here to enter text., with the fast urbanizing regions already facing great solid waste management challenges [2]. Poor waste management in developing nations leads to tremendous health and environmental burdens [3].

Meanwhile the universal need for clean and affordable energy is in play, with the depletion of fossil fuel sources and their associated negative environmental effects fueling the shift towards renewable energy sources [4]. Researchers are therefore pursuing sources which can provide reliable energy supply, that are cost-effective and carbon-free or -neutral.

Biomass waste is the 4th largest global energy source after coal, oil and natural gas, and concentrates energy first derived from the sun through photosynthesis [5]. In the context of this study, the term ''organic municipal waste'' is used to refer to forms of biomass which

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Nomenclatures

AC	Alternating Current
AD	Anaerobic Digestion
Aspen	Advanced System for Process Engineering
BOP	Balance of Plant
CRF	Capital Recovery Factor
DC	Direct Current
EU	European Union
GDP	Gross Domestic Product
GHG	Green House Gas
CAPEX	Capital Expenditure
FiT	Feed-in-Tariff
HHV	Higher Heating Value
HOMER	Hybrid Optimization of Multiple Electric Renewables
ICE	Internal Combustion Engine
IRENA	International Renewable Energy Agency
LCoE	Levelized Cost of Electricity
LFG	Land Fill Gas
LHV	Lower Heating Value
MSW	Municipal Solid Waste
NPV	Net Present Value
O&M	Operation and Maintenance
OPEX	Operational Expenditure
ORC	Organic Rankine Cycle
PV	Photovoltaic
RMSE	Root Mean Square Error
STBR	Steam to Biomass Ratio
US	United States
USD	US Dollar
VSPP	Very Small Power Plant
Wt	Weight

include a combination of wood chips, food residue, agricultural waste, fruit peels and waste as well as uncollected fraction of paper and pulp regarded as secondary biomass materials, found in landfills.

In the United States (US), the European Union (EU) and a number of Asian countries, recovery of energy from MSW has been successfully done. However, in Africa, only a limited number of successful waste-to-energy technologies have been reported. There has been success in cogeneration (Namibia, Nigeria, South Africa) and landfill gas (LFG) to electricity (Mauritius, South Africa) [6]. Cameroon has managed to recover and combust bio-methane from MSW landfills [7]. For MSW incineration, a functional plant was commissioned in Addis Ababa in 2018 [8].

Despite proliferated availability of MSW, the investment and technical requirements of waste-to-energy systems as a waste disposal strategy are the main causes for delayed adoption [9]. Moreover, most African nations lack up-to-date physicochemical characteristics data of the available MSW, and are hence rendered unready for waste-to-energy recovery [10]. The generated waste differs in composition, quantity and generation rate. This greatly depends on a country's dietary, economy and environmental conditions, with generation rate being directly proportional to income levels. Higher GDP countries waste is predominantly paper and packaging waste, while lower GDP countries generate mostly organic biodegradable waste [11].

Numerous studies have been conducted previously focusing on technologies for MSW processing using life cycle assessment methods to critique their environmental effects and economic feasibility [12]. The conclusions made over time claim that sustainable MSW management can be achieved through a combination of land filling and composting, incineration and landfill gas recovery systems, and finally by anaerobic digestion (AD) and gasification [13,14].

A research study conducted in Pakistan on gasification of solid waste and biomass for electricity generation concluded that those two renewable energy resources are suitable for power generation and gasification is feasible for all waste types processing [15]. A recent study in Thailand coupled an organic Rankine cycle (ORC) to an incinerator for infectious medical and municipal wastes to assess its power generation potential in a very small power plant (VSPP). It was found to be economically feasible with a LCoE (Levelized Cost of Electricity) of 0.257 USD/kWh [16].

A solar-rice husk DC micro-grid electricity supply to a village of 450 households has also been assessed in a technical business framework in the US [17]. The system achieved an LCoE of 2.05 USD/kWh and was reported as economically feasible. A techno-economic evaluation based on modeling and optimization of a PV-biomass micro-grid to power a decentralized apple farm was presented in a study in Saudi Arabia reporting a 0.339 USD/kWh LCoE [18].

A micro-grid system composing of agricultural waste, geothermal and solar PV with a battery backup system is presented in a study in Romania to electrify a greenhouse [19]. The study found that setting up decentralized energy systems to use locally available renewable resources as more economically feasible and environmentally sustainable than grid extension.

A number of studies have also focused on simulation of the gasification process using computation software. In a study by [20] a steam gasification system was modeled incorporating a variety of biomass feeds. The composition of syngas was evaluated for each biomass feed. Doherty W. et al [21] modeled a fluidized bed biomass gasifier and predicted the syngas composition, heating value and cold gas efficiency. Gasification temperature, biomass moisture, STBR (Steam to Biomass Ratio), air–fuel ratio and air and steam temperature were varied to study the influence on syngas composition and heating value. The study concluded that biomass moisture greatly influenced the cold gas efficiency. Adnan M. A. et al [22] modeled a novel configuration for - Spirulina microalgae gasification and investigated its performance using Aspen Plus. The effects of oxygen and steam gasification on the system's performance were investigated.

Kombe, E. Y. et al [23] developed a thermodynamic model for of sugarcane bagasse air gasification in a downdraft gasifier and Aspen Plus software to predict syngas composition at varying operating conditions. Furthermore a sensitivity analysis was performed to determine the influence of gasification temperature, moisture content and equivalence ratio on the syngas composition, yield and lower heating value. Response surface methodology was employed to optimize syngas energy content and CO_2 emission minimization.

Various studies concur that optimizing hydrogen concentration in gasification products, while integrating carbon capture is the most environmentally sustainable way to implement this waste-to-energy technology. Cao et al., 2021 [24] investigated air-gasification of pine sawdust using dolomite as the in-bed material and the influence of gasification conditions on yield properties. The findings revealed higher hydrogen production and tar conversion efficiencies with increased dolomite content. In his study Cormos, 2023 [25] concluded that green hydrogen generation from sawdust with incorporated decarbonization resulted in potentially high energy conversion efficiency of 57–59 %, avoided energy and costs from decarbonization penalties of 2.2–3.5 net points, and finally negative CO_2 emissions. Steam gasification has also been assessed in numerous studies with proven increase in H_2 content in the product gas [26–29].

Despite the many studies focusing on MSW to energy conversion, it is notable that there is a knowledge gap in the energy generating potential of the organic fraction of this waste as a whole, and the economic feasibility of implementing gasification power plants to process MSW in

Kenya. Previous works have majorly focused on waste -to-energy conversion potential for specific species of organic waste such as rice husks, sugar bagasse and heterogeneous agricultural waste singularly. However in the wake of the MSW management crisis, it has become imminent to investigate the potential of combined materials in waste, accommodating fractions of uncollected secondary biomass such as paper, pulp and cotton. Moreover, research to provide an overview of the techno-economic performance of MSW gasification micro-grids, as a basis for implementation would be beneficial in policy and decision making for investors and the government initiatives.

This study therefore presents a novel economic evaluation regarding a grid-connected micro-grid plant based on steam gasification of organic MSW. Among this work's contributions are the physico-chemical, ultimate and proximate analysis data for MSW in Nairobi, Kenya, and the waste's syngas quality which informs the power generating potential of the waste within the said location. The study's findings should therefore be available to the scientific community, and be used as fundamental information for decision-making on waste-to-energy projects.

In this study, a downward draft steam gasification power plant is modeled and simulated on Aspen Plus software, incorporating MSW ultimate and proximate analyses of data from Dandora dumpsite located in Nairobi, Kenya. A real plant is then selected from literature to cater for a selected load profile and evaluated for its economic feasibility, technical performance as well as its potential to abate grid-related GHG emissions on HOMER® Pro software. Inputs to the techno-economic analysis are thus based on the performance parameters obtained from the simulation.

2. Theoretical principles

Waste-to-energy conversion methods vary from physico-chemical, bio-chemical and thermo-chemical processes. Gasification is the process by which carbonaceous matter is thermo-chemically converted into syngas, with a higher calorific value. The feedstock is reacted with limited amounts of air and/or steam referred to as the gasification agent [30]. The feedstock is heated up for moisture removal in the drying zone, then decomposed into basic constituent elements in the pyrolysis zone before reacting with the gasification agent and with each other in the combustion and reduction zones. The input parameters that most affect the gasifier output are gasifying agent to fuel ratio, bed material, temperature, particle size and carbon content of the biomass [30–33].

The gasification process hence consists of drying, pyrolysis, combustion and reduction zones. In a drying zone the biomass is heated to reduce its moisture. It then undergoes decomposition process in the pyrolysis zone. The products of decomposition process react with each other and also with the gasification medium in the reduction zone. Drying, pyrolysis and reduction regions involve endothermic reactions which are supplied with heat from the combustion region containing exothermic reactions [34].

The subsequent reactions that take place in the gasification process are presented in Table 1. Once in the gasification system, carbonaceous material undergoes pyrolysis where char, tar and volatile substances are produced. The char containing mostly carbonaceous matter is then combusted and reduced in subsequent and simultaneous reactions resulting to producer gas which consists of carbon dioxide, hydrogen, carbon monoxide, methane and traces of nitrogen gases. A pure mixture of CO, CH_4 and H_2 is referred to as syngas, a combustible gas fuel.

Aspen Plus is a process simulation software widely used in chemical engineering and process design and whose platform enables modeling, simulating, and optimizing various processes. It has a vast library of customizable pre-built unit operation models and property databases, to suit specific applications.

Gasification models are developed on Aspen therefore enabling researchers to analyze different scenarios while saving time and avoiding costly experiments. The available models are classified into kinetics, neural network, and thermodynamic equilibrium models where the latter is useful in syngas equilibrium composition prediction by Gibbs free energy minimization [37]. Either stoichiometric or non-stoichiometric techniques can be used where the former is based on equilibrium constants of a set of reactions as function of change in the Gibbs free energy [38]. On the other hand the latter is based on minimizing the Gibbs free energy of the reacting species. The stoichiometric technique can be complex as it requires definition and description all possibly occurring reactions in a gasifier, which may not be fully known [39].

Table 1
Gasification reactions of carbonaceous materials [35,36].

Reaction	Reaction Name	ΔH25 °C (kJ/mol)
Pyrolysis		
Carbonaceous material → *Char* + *Tar* + *volatiles*	Pyrolysis reaction	
Combustion		
$C + 0.5O_2 \rightarrow CO$		−111
$C + O_2 \rightarrow CO_2$	Total combustion	−394
$CO + 0.5O_2 \rightarrow CO_2$	CO combustion	−283
Reduction	Partial combustion	
$C + CO_2 \rightarrow 2CO$	Boudouard	+172
$C + H_2O \rightarrow H_2 + CO$	Steam-carbon	+131
$C + 2\,H_2 \rightarrow CH_4$	Hydrogasification	−74.8
$CO + H_2O \rightarrow H_2 + CO_2$	Water-gas-shift	−41.2
$CO + 3H_2 \rightarrow CH_4 + H_2O$	Methanation	−206

HOMER® Pro enables users to design and analyze off-grid and grid-connected renewable energy systems by considering various energy sources, such as solar, wind, hydro, biomass, and diesel generator. It incorporates a comprehensive set of features and algorithms that allow for detailed modeling and optimization of energy systems, taking into account factors like system components, costs, energy resources, and loads. It is a powerful techno-economic evaluation tool that calculates key metrics such LCoE, NPV and payback period in order to assess the financial viability of renewable energy systems by taking into account project CAPEX, OPEX and component full lifetime.

3. Methodology

3.1. Dumpsite location and sampling criteria

Dandora dumpsite whose geographical coordinates are 1° 29′ 0″ South, 36° 54′ 0″ East, is located in Nairobi, Kenya. As Africa's 4th largest dumpsite, it is the principle dumpsite within the region covering over 30 acres of land in the heart of the Nairobi slums of Korogocho, Baba Ndogo, Mathare and Dandora.

All waste collected from Nairobi city is dumped off at the open landfill located approximately 7.5 km northeast of the city. Despite being declared full by the Nairobi council in 2001, an estimated 2,000 tons of waste are still dumped at the site each day [40]. Fig. 1 shows the geographical location of the study sites.

MSW of 50 kg was collected from different points of the dumpsite using random sampling criterion where waste was collected from each of 5 strategic accessible points namely James Gichuru Primary School, Wangu Secondary School, Dandora Health Centre, Rehoboth and Rock churches. MSW physical composition was then determined as percentages in terms of glass, plastics, rubber, textiles, agricultural waste, food residue, wood waste, and paper (pulp); which was then mainly classified as organic or inorganic waste. As illustrated in Fig. 2, the organic fraction was then isolated, sequentially mixed applying the quartering method of sampling where it was subdivided into 8 portions namely I, II, III, IV, V, VI, VII, and VIII to be classified into even (II, IV, VI, and VIII) and odd portions (I, III, V, and VII). Even amounts were mixed and again separated into two samples namely M and N. Similarly, odd sample was mixed and separated into two samples namely O and P. Then these four

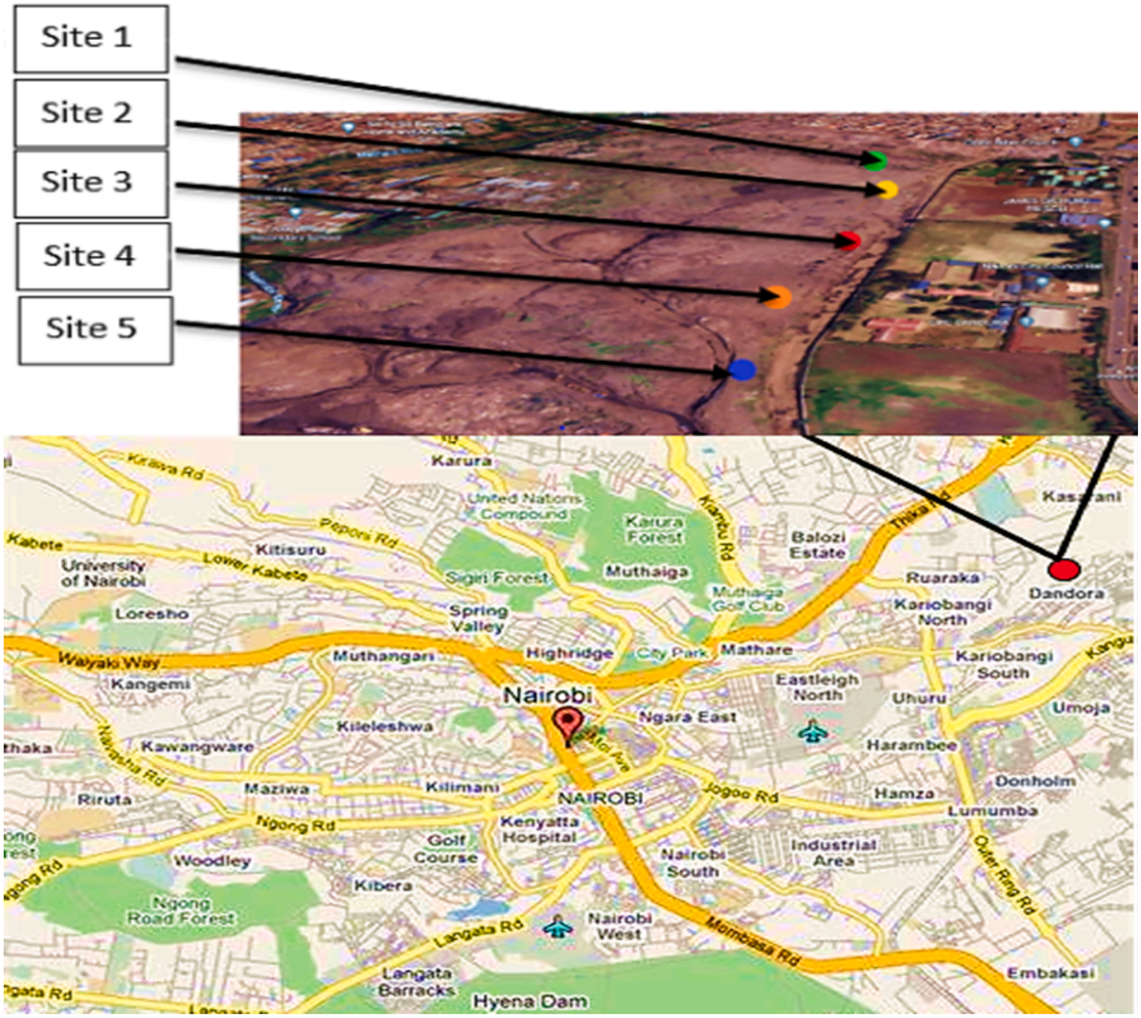

Fig. 1. Map of the geographical location of the study sites.

Fig. 2. Quartering method for waste sampling adopted from Yatsunthea & Chaiyat (2020).

samples (M, N, O and P) were mixed diagonally like M & P and N & O and separated into two samples like M & P into Y and N & O into Z. Finally, Y and Z were mixed for the final sample of analysis. A final weight of 0.5 kg was extracted and taken through laboratory proximate analysis [16].

Whereas seasonal variations may have significant effects on the type and amount of MSW generated in some regions, studies have shown a less than 7 % deviation in the physical composition and amount of MSW generated in sub-Sahara Africa based on the dry and rainy seasons experienced [41–43]. For those reasons, seasonal variations in waste generation were not considered in the study.

3.2. Simulation procedures

Aspen Plus software was used to simulate the gasification process reason being that the application is useful when dealing with non-conventional (organic waste and ash) and solid (char) substances. Its user-friendliness enables easy simulation of the gasification process simulation after model development [34].

Aspen Plus Version 10 was used to model steam gasification of MSW-Organic fraction in phases, incorporating the acquired ultimate and proximate data as inputs. The 5 phases comprised feedstock drying, conversion, pyrolysis, gasification and finally syngas purification as shown in Fig. 3.

In the drying phase, the feedstock's moisture content is lowered before being loaded into the CONVERTER for breakdown into elemental composition and later into the PYROLYZER and GASIFIER reactors where pyrolysis and gasification processes subsequently take place. The gasifier is supplied with steam from a boiler with a water inlet (INLETW). The syngas produced passes through a cyclone (ASHSEPAR) for ash removal, and is then cooled before present moisture content is extracted at the water separator (WATERSEP). The syngas is further

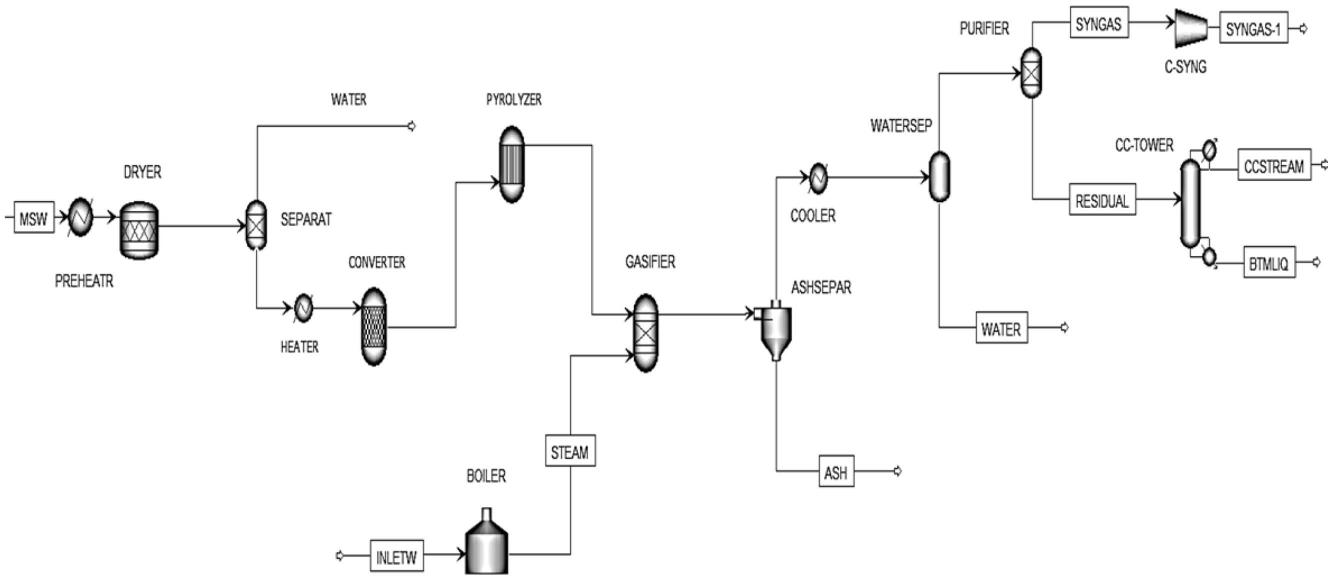

Fig. 3. Gasification process simulation flow sheet modeled on Aspen Plus.

purified from unwanted soluble SO_X and CO_2 gases, and residual tar is redirected into a carbon capture tower (CCTOWER) containing an inlet carbon capture stream (CCStream) containing the liquid for CO_2 chemical extraction, and an outlet of purified liquid tar (BTMLIQ). A syngas compressor (C-SYNG) is also integrated for efficient gas storage. The gasifier is simulated using RGibbs reactor where the simulation is based on Gibbs free energy minimization to obtain varying products using MSW as a feedstock [44]. Organic MSW was chosen as a fuel and was introduced to the software using its proximate and ultimate analyses [45].

3.2.1. Operating conditions

The operating conditions of the simulation process for the drying, elemental conversion, pyrolysis and gasification chambers as well as steam and MSW feed-in rates are shown in Table 2.

3.2.2. Performance parameters

To evaluate the gasifier's performance, a number of parameters were analyzed namely Q_{yield}, LHV_{Syngas} and gasification ratio, where Q_{yield} is the amount of produced syngas calibrated in Nm^3/kg and is a function of syngas flow rate and by the MSW flow rate as shown in Eq.(*1*). LHV_{Syngas} is the amount of heat released during complete combustion of the syngas fuel, excluding the water's heat of vaporization in the combustion product. In this case LHV_{Syngas} is a function of the individual LHVs of its constituent gases H_2, CO, and CH_4 [46] as expressed in Eq.(2):

Table 2
Operating conditions and feed stream input conditions for the process simulation [44].

Block information		Operating conditions	
Name	Type	Temperature (°C)	Pressure (atm)
DRIER	Rstoic	100	1
CONVERTER	Ryield	500	1
PYROLYZER	Rgibbs	500	1
GASIFIER	Rgibbs	700–900	1
Feed streams	Operating conditions		
	Temperature (°C)	Pressure (atm)	Flow rate (kg/hr.)
Wet MSW	25	1	100
Steam	110	1	100

$$Q_{yield} = \frac{\dot{V}_{syngas}}{\dot{m}_{MSW}} (Nm^3/kg) \tag{1}$$

$$LHV_{Syngas} = 35.81X_{CH4} + 12.63X_{CO} + 10.79X_{H2} (MJ/Nm^3) \tag{2}$$

where X_i represents the molar fraction of each syngas component on a dry basis [47].

The gasification ratio, is defined as the mass ratio of produced syngas to MSW consumed by the gasifier as expressed in Eq.(3) where ρ_{syngas} is the density of syngas and $\dot{m}_{MSW}$ the MSW mass flow rate.

$$Gasificationratio = \frac{Q_{yield} \times \rho_{syngas}}{\dot{m}_{MSW}} (kg/kg). \tag{3}$$

3.3. Assumptions for simulation process

The simulation process was considered to take place in steady state, isothermal and isobaric conditions, with complete non-conventional to conventional conversion (waste conversion to basic element constituents). All reactions were assumed to reach equilibrium conditions and reaction kinetics were not taken into account. The producer gas stream was taken to contain mainly CH_4, CO_2, CO, H_2 and H_2O, where N_2 was disregarded. The considerations that ash is inert and that organic waste devolatilization takes place instantaneously were also taken into account as also previously applied elsewhere [23,48–50].

3.4. Model validation

The developed model was tested by running a simulation using ultimate and proximate data from an experimental study done previously by [26]. The study involved steam gasification of wood residue to produce hydrogen-rich syngas. The authors used a research scale fluidized bed, with reactor temperatures of 700 °C, 800 °C, and 900 °C to obtain different levels of hydrogen composition in the syngas. The experimental parameters are shown in Table 3. The obtained simulation results were then compared to the experimental results, performed under the same operating conditions as the simulation [34,48,51].

It is noteworthy that the study used ultimate and proximate analysis data of wood residue from (Fremaux et al., 2015) which showed the most characteristic similarity for C, H, fixed carbon, HHV and volatile matter contents, to validate the model developed. This approach focused on similarity in ultimate and proximate qualities contrary to using MSW

Table 3
Data for model validation (Fremaux et al., 2015).

Ultimate analysis	(wt %)	Proximate analysis	(wt %)
C	50.26	Volatile matter	77.71
H	6.72	Fixed carbon	16.94
O	42.66	Ash	0.34
N	0.16		
S	0.20	HHV(MJ/kg)	19.97

data from literature which showed larger disparities as can be observed in previous literature [52–54]. Wood residue also shares physico-chemical similarities with the organic MSW feedstock under study, which largely constitutes biomass.

The root mean square error (RMSE) was then estimated using Eq. (4) as a measure of the model's accuracy to replicate experimental conditions and therefore results. A deviation of up to 10 % was considered within acceptable range as observed in literature [20,23,51].

$$RMSE = \sqrt{\frac{\sum_i^n (Model - Exp)^2}{n}} \tag{4}$$

where n is the number of set data points, *Model* the simulated results and *Exp* the experimental results for each gas species.

The influence of Steam to Biomass Ratio (STBR) and temperature on the syngas species was also investigated in order to conduct a sensitivity analysis of the simulation model [30,33,55]. This was performed making 0.2–1.0 STBR variations at 700 °C and temperature variations of 600 °C- 900 °C at 1.0 STBR.

3.5. Techno-economic analysis

3.5.1. Gasifier unit and component costs

A real gasification system design was adopted from [56] with modification and techno-economical assessment was performed for the micro-grid consisting of mainly a 60 kW downward draft gasifier and a 100 kW ICE coupled to a generator. The tool for techno-economic analysis used was HOMER® Pro. The rest of system components included an automatic belt conveyor used to discharge ash from the bottom of the reactor into an ash drum, a cyclone separator to eliminate ash particles from syngas, an ash collecting system equipped with a screw conveyor, and a water- acetone syngas cleaning system. The setup of the gasifier unit was as shown in Fig. 4. This design's features were a good match for the system modeled on Aspen Plus and incorporated the critical components. Based on the available real system, an electrical community –based load of 165.59 kWh, 23.31 kW peak was then selected on HOMER® Pro, for scenario analysis.

There are various parameters that greatly influence a project's economic feasibility. Among those used in this study included capital costs, total operation and maintenance (O&M) costs, organic MSW feedstock cost, plant size, plant lifetime, electricity price and feed-in tariff (FiT).

The individual equipment costs of the gasifier system are presented in Table 4 and sum up to make the total capital costs.

In this study the MSW feedstock was assumed to be freely acquired from Dandora dumpsite taking into consideration high waste availability and possible waste treatment incentives and reimbursements. Minimum wages specific to Kenya were used to account for labor payments in plant operation and MSW sorting activities. A total of 20 years of plant life was assumed, with an ICE life of 20,000 h which necessitated calculation of replacement costs.

The biodiesel consumption was calculated according to Eq.(5) to determine the fuel requirements for the steam generating boiler, providing the gasifying agent [62].

$$V = \frac{(c_w + L_V)}{LHV \times \eta_{Boiler} \times \rho_{Bd}} \tag{5}$$

where here V is biodiesel consumption in liters, C_w the specific heat capacity for water 4,186 J/kg, L_V the specific latent heat of vaporization of water 2,260 kJ/kg, LHV is the Lower heating value of biodiesel 38.5

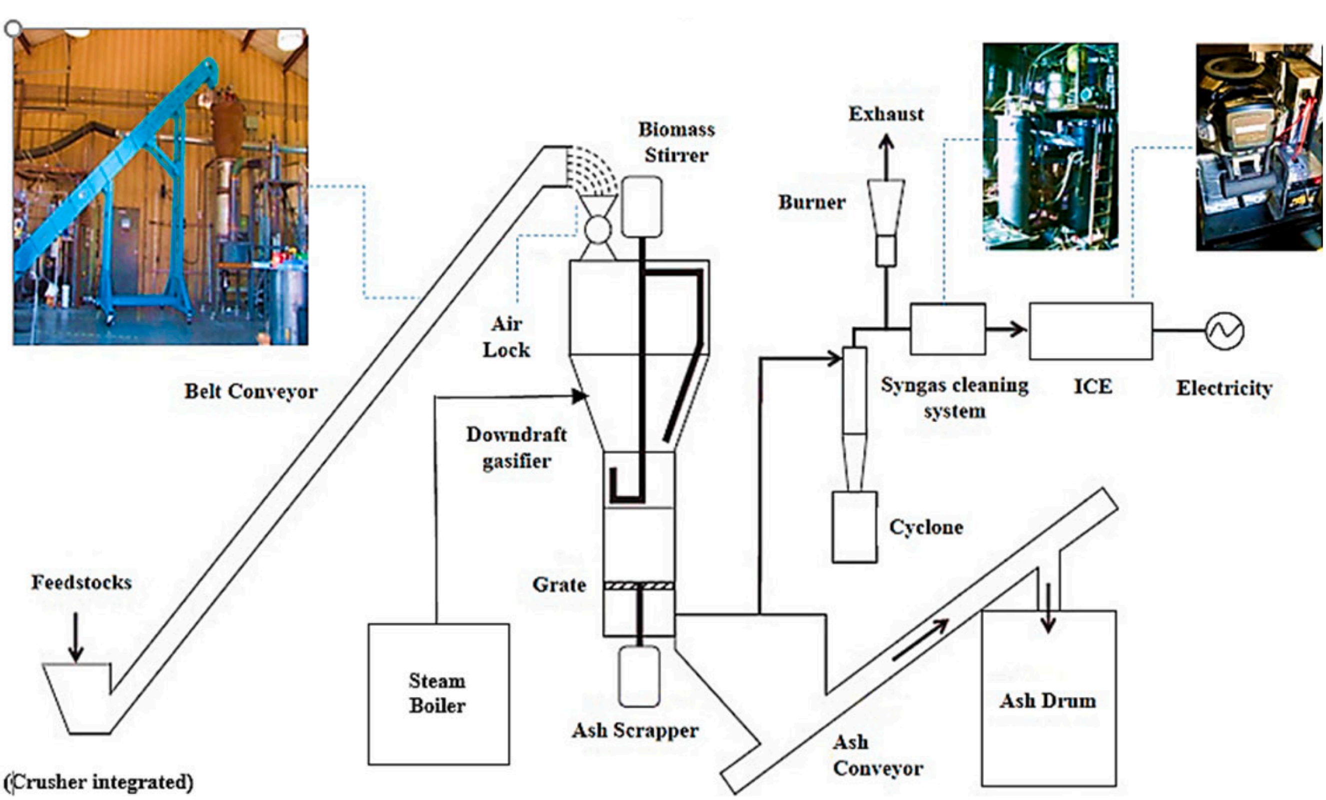

Fig. 4. Small-scale gasification plant (Indrawan et al., 2020) with design modification to incorporate steam gasification.

Table 4
Capital and O&M costs of the gasification system.

Description	Amount, $/month	References
Capital costs		
Reactor, cyclone separator, and control system	60,000	[56]
Belt conveyor and crusher	11,000	Bunting Magnetic Co.
Ash removal system (ash drum, screw conveyor, electric motor)	10,000	[56]
Steam generating boiler	10,000	Zichuan Zhungen Co. Ltd
Gas scrubbing system (double gas scrubber, pump)	4,500	The gas cleaning system consisted of water–acetone solution [57]
Power generation unit (natural gas ICE)	18,000	An ICE is used to accommodate total volume of syngas flow with an assumed capital cost based on a detailed specification in [58]
Total	113,500	
Operational costs		
Human resources		
Labor_1 plant operator	656	
Sorting 35$/day man power	1050	
Electricity for BOP		
Electric heaters, 5 pcs @360 W_1314 kWh/month	192	
Chiller, 1.5 hp 815.8 kWh/month	120	Schreiber, Model 300 AC, Engineering Corporation, Cerritos, CA, USA
Water pump, 0.5 hp_543.9 kWh/month	80	
Belt conveyor, 1 hp_543.9 kWh/month	80	(Bunting Magnetics Co., energy
Crusher_1440 kWh/month	211	(Bunting Magnetics Co.)
Ash scrapper, 1 hp_543.9 kWh/month	80	(Dayton, Model 2MX14A, Dayton electric Mfg.Co., Lake Forest, IL., USA)
Ash scrapper, 1 hp_543.9 kWh/month	80	(Dayton, Model 2MX14A, Dayton electric Mfg.Co., Lake Forest, IL., USA)
Syngas cleaning system (i.e., acetone)_5 gal/day, $11.48/gal	1722	Water-acetone based, mixable with renewable filters [59]
Disposal cost of liquid waste (i.e., acetone)_5 gal/day, $0.23/lb. & density 784 kg/m^3	225	Hazardous Materials Management Facility, [60]
Biodiesel_28.195 Gal/day, $5.44/gal	4602	
Maintenance costs [56]		
Tools	25	
Sealant and insulations	20	
Air lock fins, 8pcs	200	
Spare electric motor	17	
Total O&M costs $/month	9360	
Calculated using electricity rate of $0.146/kWh [61]		

MJ/kg and ρ_{Bd} the density of biodiesel. A boiler efficiency (η_{Boiler}) of 90 % was assumed.

3.5.2. Economic feasibility determinants

The load is assumed to be solely supplied by the gasification power plant, and the extra generation fed into the utility grid. The balance of plant components electricity demand was calculated using a power purchase tariff for businesses in Kenya of 0.146 $/kWh. The FiT of 0.95 $/kWh specific to Kenyan plant capacities of 20 MW and below was considered for the case study, to sell back the excess electricity to the utility grid. The discount rate of 8.75 % and inflation rate of 7.38 % were also held in regard as actual economic input data from the Central Bank of Kenya [63].

The NPV and LCoE were the two major investment appraisal and capital budgeting decision tools used to determine the project's feasibility [56,64] as expressed in Eqs. (6), (7), Fehler! Verweisquelle konnte nicht gefunden werden.), Fehler! Verweisquelle konnte nicht gefunden werden.) and (8):

$$NPV = \sum_{t=0}^{n} \frac{CF_t}{(1+k)^t}; \tag{6}$$

$$LCoE = \frac{Netannualcostofsystem(\$/yr)}{Usefulelectricenergyproduced(kWh/yr)}; \tag{7}$$

where CF is cash flow ($), k the % annual interest rate, t the corresponding year; n the project duration. CF is also termed as the net annual cost(annual cost minus revenue) [56]. The expression (1) $+ k)^t$ is the capital recovery factor (CRF) converting present value in a sequence of equivalent annual cash flow over a project period. The LCoE refers to the average cost for generating a unit of energy as shown in Eq.(7) [64].

The NPV method takes the time value of money into consideration, and is computed using real discount rates computed based on the actual available inflation rate as shown in Eq.(8). In this study, the annual interest rate *(k)* was equated to the real discount rate (r_r) to determine the project's NPV.

$$r_r = \left\{\frac{1+r_n}{1+I}\right\} - 1 \tag{8}$$

where r_r is the real discount rate,r_n the nominal discount rate (8.75 %) and I (7.38 %) the inflation rate as provided by the publication of Central Bank of Kenya [63].

3.5.3. HOMER® pro system design

HOMER® Pro is particularly suited for practical micro-grid systems modeling and was used as the primary micro-grid analysis tool. The methodology, adopted from [65] allows one to take advantage of the inbuilt biogas power plant modules in HOMER® Pro. Technical and economic parameters were edited and completed to simulate an ICE fueled with syngas produced by a downdraft gasifier (gasification system). The micro-grid gasification system configuration on the software is illustrated on Fig. 5.

Through the AC bus, electricity can be channeled from the power plant into the grid and the load suppled with electricity solely from the plant. The economic feasibility of the organic MSW gasification plant was analyzed through profitability analysis. The specifications related to technical and economic parameters indicated in Table 5 were used as inputs to the modeled system for analysis on HOMER® Pro. A zero capacity shortage was assumed as the power system capacity is well above the load demand and is able to meet this target throughout the project lifetime.

4. Results and discussion

4.1. Waste characterization and analysis

Physical characterization, ultimate and proximate analyses results obtained for the MSW collected from Dandora Dumpsite in Nairobi

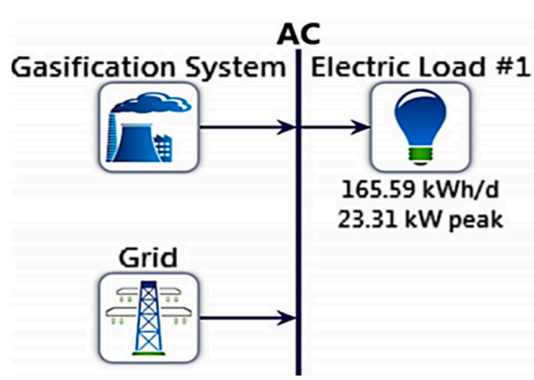

Fig. 5. Micro-grid system configuration on HOMER® Pro.

Table 5
Key economic inputs for HOMER® Pro modeling and economic analysis.

No	Parameters	Value
A	Feed rate/Capacity, tpd	2.4
B	Total output power, kW	60
C	Capacity shortage, %	0
D	Total capital cost	113,500
E	Total O&M costs \$/yr.	112,320
F	Project (economic) life, years	20
G	Electricity price, \$/kWh	0.146
H	FiT, \$/kWh	0.095
I	Gasification ratio	2.2705
J	LHV_{Syngas}, MJ/kg	24.26

County, Kenya are presented in

Fig. 6. It was observed that organic waste was the greatest constituent at 64.2 %, which included uncollected paper (pulp) as a secondary organic material. From.

Fig. 6, inorganic waste registered 35.8 % of MSW constituent. This is in agreement with previous results which showed organic waste composition of over 60 % in Kenya's MSW [41].

The proximate and ultimate analyses presented in Table 6 revealed a high volatile matter content of 76.10 %, as well as high carbon content of 50.04 % in the organic MSW and was in agreeable range with a study on wood chips [21]. This is in consideration of the MSW sample being rich in biomass content.

4.2. Model validation results

Computational results of wood residue were compared to those obtained in an experimental study reported previously [26]. The simulation was run under conditions of 1 bar atmospheric pressure, 700 °C temperature, 2.5 kg/hr. flow rate, steam to biomass ratio (STBR) of 1.0 and a particle size distribution of 1–2.5 mm. The comparison of the experimental and simulation results were as shown in Fig. 7.

Observe from Fig. 7 that, the model is able to adequately predict the concentrations of the constituent gas species in the producer gas. A good agreement between the experimental results and simulated results for CO_2 fractions is observed. There is a small deviation in the CO concentration as the model does not accurately depict the true phenomenological behavior of steam MSW gasification as in an experimental setup. There is a difference in methane and hydrogen yields, where the computational CH_4 value is significantly underestimated while that of

Table 6
Proximate and Ultimate analysis results.

Ultimate analysis	(wt%, dry)	Proximate analysis	(wt%, dry)
C	50.04	Volatile matter	76.10
H	7.12	Fixed carbon	17.03
O	32.85	Ash	6.87
N	1.49		
S	1.40	Calorific value (MJ/kg)	19.64

H_2 is moderately overestimated. This is attributable to the equilibrium model, which yields almost complete CH_4 conversion contributing to hydrogen formation whereas actual gasifiers, which due to their short residence time, are unable to achieve thermodynamic equilibrium. As a result, much lower CH_4, and higher H_2 values are quite common in simulations with equilibrium modeling [51].

The model's calculated RMSE value is 6.7 % as illustrated in Eq.(4) which was found to be quite acceptable. Compared to a 1207.68 kJ/Nm^3 LHV from the experimental data, the simulation resulted in a LHV of 1003.12 kJ/Nm$^{3.}$ This deviation in syngas energy content can be accounted for by the differences in gas species concentrations, each of which have their own LHVs.

4.3. Obtained syngas performance parameters

The model was further used, integrating the current study's ultimate and proximate data analyses at 700 °C and steam to biomass ratio (STBR) 1.0 operating conditions. The results revealed a syngas yield of 2.39 m^3/kg, with gas species molar fractions of 61.15 % H_2, 24.12 % CO, 13.30 % CO_2 and 1.42 % CH_4. A LHV of 1015 kJ/Nm^3, an equivalent of 24.26 MJ/kg was therefore calculated.

Negligible methane production is due to thermodynamic equilibrium state achievement during the simulation. The high hydrogen content can be attributed to steam gasification, generating more hydrogen from steam-methane reforming process and the water–gas reaction, compared to air gasification as in agreement with [66]. Syngas output parameters obtained showed a 23.8 % higher energy content compared that of raw MSW as a fuel (19.64 MJ/kg). It was, therefore, argued that gasification is a more energy efficient processing technology than incineration which utilizes the waste in its raw form.

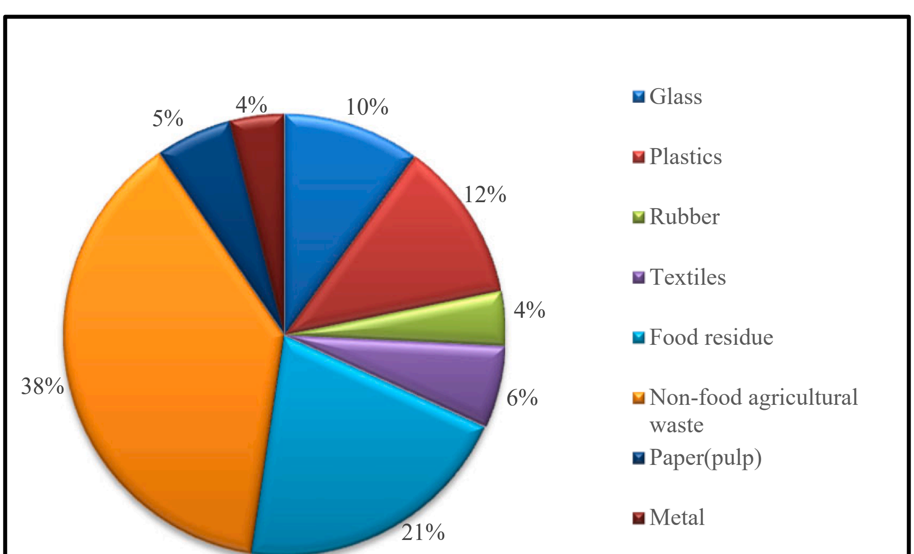

Fig. 6. MSW physical characterization according to material from Dandora, Kenya.

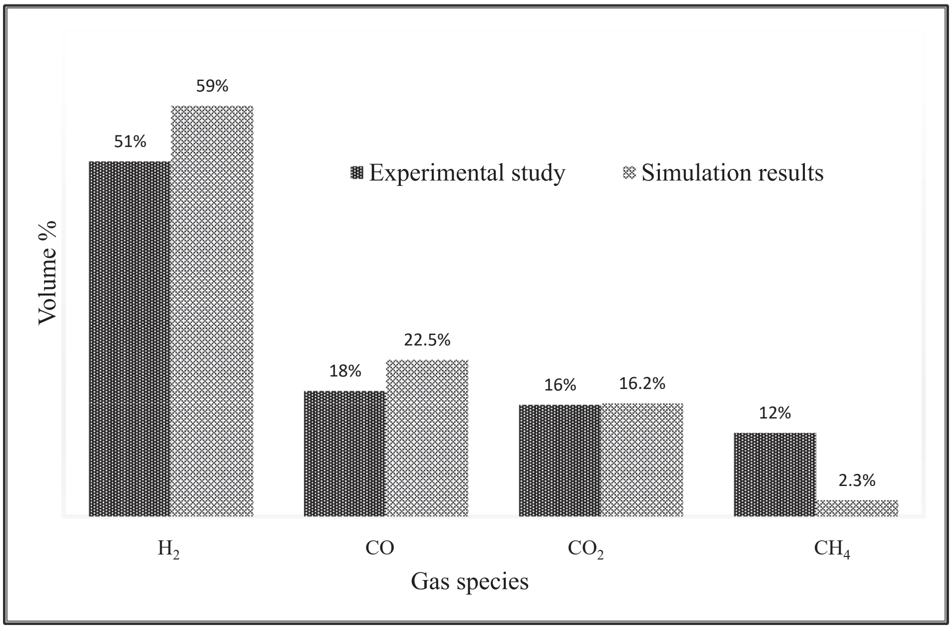

Fig. 7. Comparison of experimental and simulation results for model validation.

4.4. Sensitivity analysis

The influence of Steam to Biomass Ratio (STBR) and temperature on the syngas species were also investigated in order to conduct a sensitivity analysis of the simulation model.

The sensitivity analysis results in Fig. 8 (a) showed increasing trends in CO_2 and H_2 syngas constituents from 0.4 to 1.0 STBR, at a constant temperature of 900 °C. A decrease in CO levels was observed from 0.4 to 1.0 STBR while CH_4 values remained negligibly affected and were very low. At low STBRs between 0.2 and –0.4, the inverse phenomenon was observed where CO increased and H_2 decreased with increasing STBR, showing good agreement with previous studies [44,55,67].

A temperature variation from 600 °C to 900 °C at STBR 1 leads to an increasing trend in CO and H_2, and a decreasing trend in CO_2 and CH_4. These trends are shown in (b) and are in good agreement with experimental and computational results in literature [26,33,44].

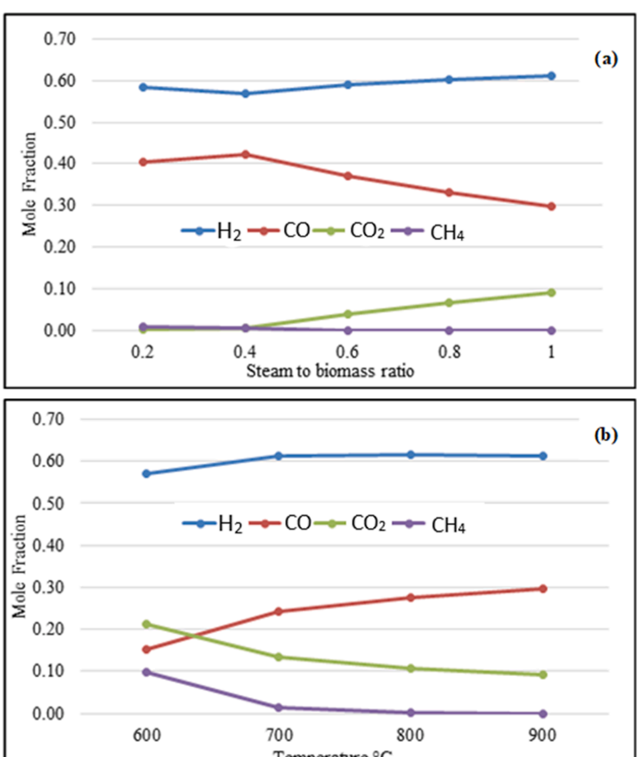

Fig. 8. Sensitivity analysis results on the influence of gasification temperature and STBR on syngas composition for MSW. (a), with variation of STBR at 900 °C, and (b), at different temperatures STBR = 1.0.

4.5. Techno-economic evaluation

The designed gasification plant was then evaluated for economic feasibility, electricity generation and carbon emissions. From Table 4, the share of operation costs is distributed as shown in Fig. 9. It is deducible that 49 % of the gasifier system operating costs are incurred by biodiesel fuel requirement of the boiler, with 21 % of the costs credited to syngas cleaning system and waste disposal of the water–acetone based syngas purification system. It can be seen from Fig. 9 that 18 % thereof is dedicated to remuneration of personnel for waste sorting and plant operation, 9 % for balance of plant equipment and 3 % credited to contingency maintenance tools.

The HOMER® Pro analytical report outputs are displayed in Fig. 10,

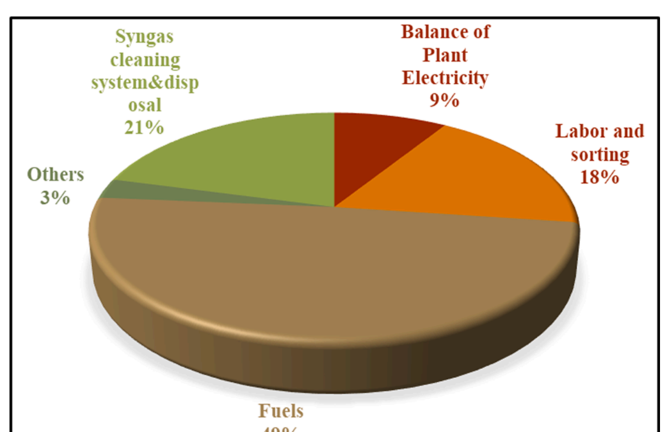

Fig. 9. Major operational costs of the downdraft gasifier system.

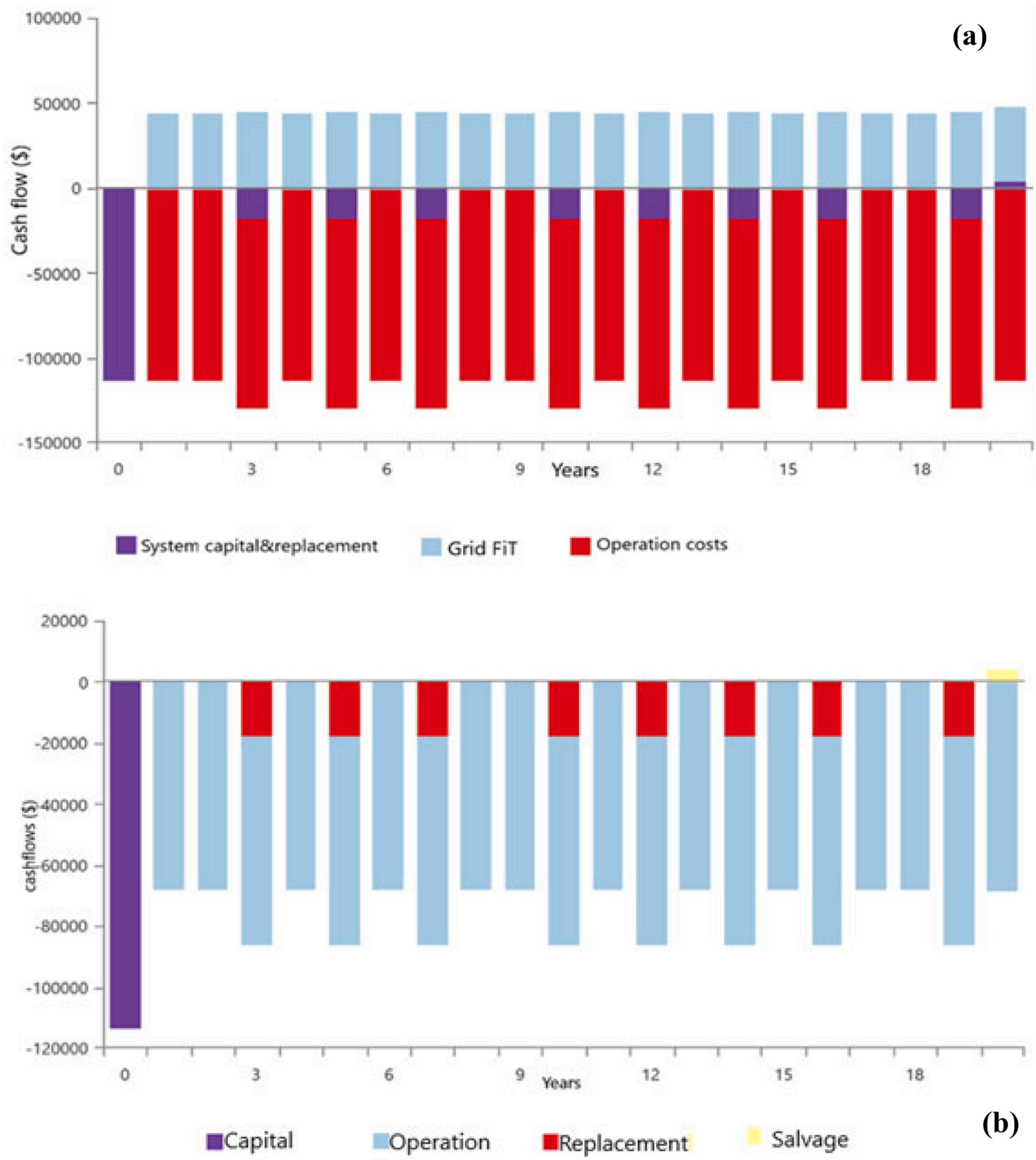

Fig. 10. Comparative costs of the gasification system in $ generated on HOMER® Pro. (a), with grid FiT, operation, capital and replacement costs, and (b), with integrated FiT and its effect on operation costs.

where the cash flows results show that overall operational costs are the greatest over the system's lifetime, compared to initial capital investment and ICE replacement costs. However, it is noted that the FiT cash inflows obtained from the grid feeding at 11.5 % self-electricity consumption and 88.5 % sale (Table 7), offsets the system's annual costs significantly by 39.34 % as deduced in Fig. 10.

Against an available MSW feed of 2.4 t/d, 0.194 t/d (8 %) is consumed to provide the required 18 kg/hr. syngas output. The gasification system costs, generation and consumption summary results are summarized on Table 7.

The annuitized project costs as well as the full lifetime NPV computations are displayed in Table. The system capital costs cover 7.9 %, operational costs 83.9 %, and salvage value 0.23 % of the total sum of project life cost.

The performed GHG emission analysis revealed negligible traces of carbon monoxide (1.16 kg/yr.), unburned hydrocarbons, (0.051 kg/yr.) particulate matter (0.00694 kg/yr.), sulfur dioxide (0.0 kg/yr.), and nitrogen oxides (1.09 kg/yr.). From the predictions, the project would produce 0.181 g/kWh of CO_2 which is a 99.8 % carbon footprint mitigation with reference to Kenya's national grid with 116 g/ kWh. Given that a mature tree absorbs up to 21 kg CO_2 /yr. on average [68], an equivalent of approximately 5 trees of average size would be used to mitigate the system's 95.3 kg/yr. CO_2 emissions. Furthermore, since a 70.8 t/yr. MSW feedstock results in 95.3 kg/yr. CO_2 emissions, it is deducible that a kilogram of MSW has a carbon footprint of 1.3 g.

The economic metrics results indicated an LCoE computed at 0.155

Table 7
Project costs, generation and consumption summary.

Consumption summary			Value			
AC Primary Load			60,442 kWh/yr.			
Grid sales			465,158 kWh/yr.			
Total energy generation			525,600 kWh/yr.			
Syngas consumption			18 kg/hr.			
Specific consumption			0.306 kg/kWh			
Mean electrical efficiency			48.5 %			
MSW feedstock requirement			8.09 kg/hr.			
Project annuitized costs						
Name	Capital	Operating	Replacements	Salvage	Resource	Total
Gasification system	$6,466	$112,320	$7,217	-$190.98	_	$125,812
Grid feed	_	-$44,190	_	_	_	-$44,190
Total system	$6,466	$68,130	$7,217	-$190.98	_	$81,622
Project Net present costs over project lifetime						
Gasification system	$113,500	$1.97 M	$126,692	-$3,352	_	$2,21 M
Grid feed	_	-$775,716	_	_	_	-$775,716
Total system	$113,500	$1.20 M	$126,692	-$3,352	_	$1.43 M

$/kWh and an NPV of $1.43 M. The obtained LCoE is 25 % less than household electricity purchase price at 0.206 $/kWh and 6 % higher than business electricity purchase price at 0.146 $/kWh in Kenya [61]. For offshore wind, solar PV and geothermal, the global weighted-average LCoE of newly commissioned projects in 2020 are; 0.084 $/kWh, 0.057 $/kWh and 0.071 $/kWh on GW scales [69]. These are comparatively 54 % for offshore wind, 37 % for solar PV and 46 % for geothermal in percentage of the study's obtained LCoE. A 60 GW bio-energy plant's global weighted-average LCoE for power projects was estimated at 0.076 $/kWh in 2020, which is 50 % of the value obtained in this study.

An economic analytical comparison was made on HOMER® Pro between the proposed gasification system with grid sales and a stand-alone plant in order to determine which option is most feasible, as shown in Table 8.

Due to the high share of energy fed into the grid, the system proved to be more economically feasible compared to a stand-alone gasification plant alternative which indicated a high LCoE and NPV of 2.08 $/kWh and $2.21 M respectively. The carbon footprint in both cases was found to be similar due to same plant capacity and feedstock considerations.

The high LCoE value can be credited to steam gasification biodiesel and boiler related costs, considering that biodiesel costs for steam boiler constituted 49 % of total operational costs. A lower LHV for syngas yield at the expense of less syngas carbon content at selected temperature 700 °C and STBR 1, have negligible effect on power generated as this is compensated by the abundant MSW feedstock. The use of ICE although a cheaper option to a gas turbine, has a short lifetime of 20,000 hrs. This necessitates at least 7–8 times replacement financial burden over the project lifetime, resulting in the high LCoE. Given that the system design was adopted from [56], the oversized Genset by a factor of 1.4 on the gasification system's capacity may have contributed to the slightly high LCoE.

The study's 60 kW gasification system is, however, considered economically viable considering the economy of scale. As a small-scale plant, it is acceptable to have a 50 % higher LCoE than a 60 GW bioenergy installation plant [69].

Table 8
Techno-economic metrics comparison between the grid connected and stand-alone plant options.

Techno- economic Metrics	Grid connected
NPV	$1.43 M
LCoE	0.155 $/kWh
CO_2	0.181 g/kWh
Techno- economic Metrics	Stand alone
NPV	$2.21 M
LCoE	2.08 $/kWh
CO_2	0.181 g/kWh

The LCoE methodology used ascribes future costs to the present value, resulting in a present price of USD 0.155 per unit energy value of 1 kWh based on an assumed lifetime of 20 years. This value is 25 % lower than the price of electricity in households, and 6 % higher than the charge for business enterprises. Gasification plant micro-grids with partial grid-feeding would therefore be economically feasible for domestic applications, offering a better trade off than grid consumption. The capital costs for such systems are however very high for average households to afford, the reason why captive power plant applications developed by consumers for self-consumption would only be viable on community-based investments, generation and distribution scales.

However, it is noteworthy that the LCoE method of economic evaluation adopted oversimplifies the project's context and cost as it ignores project risks and doesn't consider all costs including legal and land costs, critical for financial decision making.

5. Conclusion

In this paper, the physico-chemical characterization of MSW sampled from Dandora dumpsite revealed an organic waste composition of 64.12 % with high volatile matter content of 76.10 %, as well as a fixed carbon content of 17.03 %. The developed model on Aspen plus software was able to predict well the gasifier performance for different operating conditions and parameters such as temperature, and STBR, where syngas composition at 700 °C and STBR 1.0 yielded H_2 & CO as the dominant constituents (61.15 %, 24.12 %). The project was predicted to have a carbon footprint 0.181 g/kWh of CO_2 with other GHG species being negligible. Gasification was concluded to be a more efficient energy conversion technology than incineration as the syngas energy content obtained had an energy content 23.8 % higher than that of raw waste. The small-scale steam gasification plant was found to be economically feasible at LCoE of 0.155 $/kWh, given the economy of scale. Obtained values show that the small-scale steam organic MSW gasification is not only technologically, environmentally feasible but also economically viable.

The study presents a novel economic evaluation of a grid-connected micro-grid plant based on steam gasification of organic MSW. Among this work's contributions are the physico-chemical, ultimate and proximate analysis data for MSW in Nairobi, Kenya, and the produced syngas quality which informed the power generating potential of the waste within the study location. The findings are therefore available to the scientific community and project implementers for use as fundamental information for decision-making on waste-to-energy projects. The study however did not include aspects such a project risks, legal and land costs which are also critical for financial decision making. Future work can be extended to incorporate valid financing options as well as legal costs for land and licensing for higher precision in economic analysis overview. Furthermore, MSW is highly sensitive to the geographical environment and these study findings specifically apply and are limited to Nairobi, Kenya. The extension of use of the study data can apply to cities sharing similarities in the levels of industrialization, population, GDP and climate as the city of Nairobi.

CRediT authorship contribution statement

Lydia Muriuki: Conceptualization, Investigation, Formal analysis, Validation, Methodology, Writing – original draft. **David Wafula Wekesa:** Data curation, Project administration, Writing – review & editing, Supervision. **Francis Njoka:** Data curation, Visualization, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [David Wafula Wekesa reports administrative support was provided by Multimedia University of Kenya. David Wafula Wekesa reports a relationship with Multimedia University of Kenya that includes: employment. David Wafula Wekesa has patent N/A pending to N/A. None].

Data availability

Data will be made available on request.

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